MILITARY SPECIFICATION

TESTER, LENS, IMPACT RESISTANCE, DROP BALL TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a test device for determining the impact resistance of ophthalmic lenses.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| PPP-B-601 | Box, Wood, Cleated-Plywood |
| PPP-B-621 | Box, Wood, Nailed and Lock-Corner |
| PPP-B-636 | Box, Shipping, Fiberboard |

MILITARY

| | |
|---|---|
| MIL-P-116 | Preservation, Methods of |
| MIL-L-10547 | Liners, Case, and Sheet Overwrap, Water-vaporproof or Waterproof, Flexible |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-129 | Marking for Shipment and Storage |

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6540

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

DESIGNATIONS

| | |
|---|---|
| D 412-68 | Tension Testing of Vulcanized Rubber |
| D 1415-68 | International Hardness of Vulcanized Rubber |

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

3. REQUIREMENTS

3.1 Materials. Materials used for the tester shall conform to the requirements herein. Materials not specifically designated shall be of the highest commercial quality and suitable for the intended purpose.

3.1.1 Base. The base of the tester shall be cast iron or steel.

3.1.2 Guide tube. The guide tube shall be cold rolled steel.

3.1.2.1 Guide tube Support. The guide tube support shall be anodized aluminum.

3.1.3 Lens support tube and door. The lens support tube and door shall be clear rigid acrylic plastic. As an alternate the tube shall be steel.

3.1.4 Gasket. The gasket for the lens support tube shall be neoprene rubber with a hardness of 40 ± 5 durometer, a minimum tensile strength of 1,200 pounds and a minimum elongation of 400 percent when determined as specified in 4.4.1.

3.1.5 Test ball. The ball used for dropping on the lens shall be a finished carbon steel ball conforming to Grade 1000 of the Anti-Friction Bearing Manufacturers Association, Section 10 for balls. As an alternate, a stainless steel ball, grade 200 AFBMA, shall be used.

3.1.6 Hardware. Hardware items used to assemble and secure components into a sturdy unit shall be entirely suitable for the purpose intended.

3.2 Construction. The tester shall be constructed in such a way as to allow a 5/8 inch diameter steel ball weighing 0.56 ounce to fall freely from a height of 50 inches onto an ophthalmic lens being tested for impact resistance. The testing unit shall consist of a base, a guide



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tube and supports and a test area at bottom of the guide tube containing a lens support tube with gasket (as part of test block) shielded by a clear plastic door. The tester shall be furnished with a minimum of 3 test balls. The lens test block support and the base plate shall conform to figures 1 and 2. The tester shall be capable of being used to perform the test described in 4.4.1 and conform to FDA requirements.

3.3 Identification marking. Each tester shall be clearly and legibly marked with the name or trademark of the manufacturer and the contract number.

3.4 Workmanship. Workmanship shall be first class throughout. Metal parts shall be free of burrs and sharp edges. Testers shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, and investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.3 Certificates of quality. Certificates of quality supplied by the manufacturer of the tester, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the manufacturer or supplier, the contractor, the contract number, and the name of the material or component.

4.2 Sampling. Sampling for inspection (examination and test) shall be conducted in accordance with MIL-STD-105, inspection level II. Unit of product shall be one tester.



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4.3 Examination. The tester shall be examined for defects including, but not limited to, those listed in table I. The acceptable quality levels (AQLs) shall be 1.0 for Major and 4.0 for Minor defects, expressed as percent defective.

TABLE I. Categories and defects

| | |
|---|---|
| Major | |
| 101 | Component missing or of improper design |
| 102 | Tester of improper height |
| 103 | Materials not as specified |
| 104 | Test balls missing |
| Minor | |
| 201 | Metal parts not free of burrs and sharp edges |
| 202 | Identification marking incomplete, illegible or missing |
| 203 | One or two test balls missing |

4.3.1 Examination of Preparation for Delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in table II. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The Inspection Level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

TABLE II. Examination of Preparation for Delivery

| Examination | Defect |
|---|---|
| Containers | Not as specified. |
| Quantity | Quantity in interior packages and shipping containers not as specified. |
| Packaging and packing materials | Any component missing or damaged. |
| Workmanship | Inadequate application of components such as incomplete closure of container flaps, tape improperly applied, or containers distorted. |
| Markings | Omitted, incorrect, illegible, improper size, location, sequence, or method of application. |



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4.4 Tests. Tests shall be conducted as necessary to determine compliance with specification requirements. Sample size shall be as indicated in 4.2 and acceptance number shall be zero.

4.4.1 Impact test. The impact test specified in 3.2 shall be conducted as specified below: A 5/8-inch steel ball weighing approximately 0.56 ounce shall be dropped from a height of 50 inches upon the horizontal upper surface of an ophthalmic lens. The ball shall strike within a 5/8- inch diameter circle located at the geometric center of the lens. The test shall be repeated five times using different lenses. A piece of carbon paper or other suitable method shall be utilized to determine that the ball hits within the 5/8-inch circle. The ball shall be guided, but not restricted, in its fall by being dropped through a tube extending to within approximately 4 inches of the lens. The test shall be conducted with the lens supported by a tube (1-inch inside diameter, 1-1/4 inch outside diameter, and approximately 1-inch high) affixed to a rigid iron or steel base plate. The total weight of the base plate and its rigidly attached fixtures shall be not less than 27 pounds. The tube shall be made of rigid acrylic plastic, steel, or other suitable substance and shall have on the top edge a 1/8 - by 1/8 inch neoprene gasket having a hardness of 40 ± 5 durometer, as determined by ASTM Method D 1415; a minimum tensile strength of 1,200 pounds, as determined by ASTM Method D 412; and a minimum ultimate elongation of 400 percent, as determined by ASTM Method D 412. The diameter and/or contour of the lens support shall be modified as necessary so that the 1/8-by 1/8-inch neoprene gasket supports the lens at its periphery.

## 5. PACKAGING

5.1 Packaging requirements. Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit package. Each tester shall be packaged in a box designed for a type 3 load and constructed in accordance with PPP-B-601, overseas type; PPP-B-621, class 2 or PPP-B-636, class weather resistant, grade V3c. Loose components shall be packaged separately. Box shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of, or damage to the contents. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.1.1.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproof as specified in the appendix of PPP-B-636.



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5.1.2 Level B.

5.1.2.1 Unit package. Each tester shall be packaged as specified in 5.1.1 except domestic type boxes shall be used and case liner and strapping shall not be required.

5.1.3 Level C. Each tester shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.2 Packing. No further packing is required.

5.3 Marking.

5.3.1 Unit package. Each unit package shall be marked as required for the exterior container specified in MIL-STD-129.

6. NOTES

6.1 Intended use. The drop ball tester covered by this specification is intended for use in conducting the ophthalmic lens impact test required by Food and Drug Administration policy.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of packaging.
(c) National Stock Number (NSN)

6.3 This specification is intended to cover the following item listed in the Federal Supply Catalog:

| National Stock Number | Item Identification |
|---|---|
| 6540-00-083-6554 | Tester, Lens, Impact Resistance, Drop Ball Type |

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing Activity:
Defense Supply Agency - DM

Project No. 6540-1041
CS



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OMB Approval
No. 22-R255

INSTRUCTIONS. The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL T 37317

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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